Power and Cosine Filtering for Extreme Singular-Value Estimation: An Assumption-Aware Technical Reassessment of a QSVT Proposal

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Abstract

We reassess an unpublished proposal for quantum maximum/minimum singular-value estimation, focusing on its cosine-based triangular iteration. For a rectangular matrix A, we formulate the method on the normalized right Gram operator $G = (A/\alpha)^\dagger(A/\alpha)$, where $\alpha \geq \|A\|$ is an available normalization bound. A bounded even polynomial approximation to $\cos(\beta G)$ is a candidate target for a polynomial eigenvalue transformation, subject to an explicit QSP completion and normalization convention. We prove a finite-spectrum filtering bound: if the input has weight $a > 0$ on the minimum-eigenvalue eigenspace and the largest non-target filter response is at most $r < 1$ times the target response, then the conditional leakage is at most $(1-a)r^{2t}/a$ after t iterations. A Bernoulli block-encoding readout estimates the filtered Gram expectation and converts its additive error into a singular-value error bound. The result depends on target overlap, spectral separation, postselection success, normalization, and readout precision; it is not a circuit-level or end-to-end complexity theorem. Small matrix-level simulations reproduce the predicted convergence and its failures for zero overlap, narrow gaps, and right kernels. We also compare the original $\cos(\beta(\sigma/\alpha)^2)$ response with a distinct dilation-based $\cos(\beta\sigma/\alpha)$ variant. No QSVT phases or quantum circuits are synthesized, and no quantum speedup or novelty claim is made.

1 Introduction

Extreme singular values arise in numerical linear algebra and in spectral questions connected to ground-state energy estimation. An earlier manuscript, *Quantum maximum/minimum singular-value estimation algorithms* (author and publication metadata not stated in the source file; hereafter QMSVE) proposes quantum power iteration for the largest singular value, an inverse-iteration baseline for the smallest, and a cosine-based triangular iteration that avoids explicitly applying a matrix inverse [1]. This report makes that proposal precise on a right-space operator, records the conditions needed for convergence, and checks a small set of finite-dimensional examples.

The main correction is about the operator being transformed. For a general rectangular matrix, applying $\cos(A)$ is not a valid singular-value prescription. We instead set $B = A/\alpha$ and $G = B^\dagger B$ on the right singular-vector space. The QMSVE cosine rule is then $C = \cos(\beta G)$, whose eigenvalues are $\cos(\beta\sigma_i^2/\alpha^2)$. This function is positive and decreasing on the normalized spectrum when $0 < \beta < \pi/2$. A Hermitian-dilation construction gives a different response, $\cos(\beta\sigma_i/\alpha)$; we keep it separate.

Our contribution is a careful statement and finite-spectrum proof of conditional filter convergence, its measurement-to-energy conversion, and a reproducible matrix-level audit. It does not establish that the original proposal is new, that its full resource estimate is correct, or that it

outperforms existing algorithms. In particular, a gap and nonzero initial overlap remain essential, while each successful filtered sample may require many fresh attempts.

2 Related work and scope

Quantum singular-value transformation (QSVT) implements bounded polynomial transformations of singular values of a block-encoded matrix, and the Hermitian specialization implements polynomial eigenvalue transformations [2]. It supplies a possible implementation primitive for the polynomial cosine filters considered here; the present work does not synthesize its phase angles. Quantum power methods and quantum inverse iteration already have substantial prior literature [3, 4, 5]. These works differ in input and implementation models, but they rule out treating power iteration or QSVT-based powers as new in themselves.

The QMSVE manuscript is an unpublished source document with no author or venue metadata in the copy examined. Its triangular iteration is marked provisional. We therefore treat it as the origin of the algorithmic idea being reassessed, not as a verified theorem or a published result. The author's separate QCG work is cited only as prior project context and is not conflated with this method [6, 7].

3 Model and algorithm

Let $A \in \mathbb{C}^{m\times n}$, $m \geq n$, and choose a known $\alpha \geq \|A\|_2 > 0$. Define

$$B = A/\alpha, \qquad G = B^\dagger B, \qquad 0 \leq \lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_{n-1} \leq 1,$$

where $\lambda_i = \sigma_i(A)^2/\alpha^2$ are the eigenvalues of G in increasing order. The target is the smallest singular value over the full right space, including zero modes. Estimating the smallest nonzero singular value requires an additional promise or access to the kernel-orthogonal subspace. For wide matrices ($m < n$), the right Gram operator necessarily has a kernel, so the same distinction applies.

Let $|\psi\rangle = \sum_i c_i|v_i\rangle$ be a normalized input state in a right singular basis and set $p_i = |c_i|^2$. For $0 < \beta < \pi/2$, the ideal cosine-Gram filter is

$$C = \cos(\beta G), \qquad q_i = \cos(\beta\lambda_i) \in (0, 1].$$

The unnormalized state after t iterations is $C^t|\psi\rangle$, its ideal success probability is

$$Z_t = \|C^t|\psi\rangle\|^2 = \sum_i p_i q_i^{2t},$$

and the normalized accepted state's spectral weights are $p_i q_i^{2t}/Z_t$. This is the QMSVE rule applied to the right Gram operator, with normalization made explicit.

Polynomial implementation model. For an integer $m_0 \geq 0$, let

$$P_{m_0}(x) = \sum_{j=0}^{m_0} \frac{(-1)^j(\beta x)^{2j}}{(2j)!}, \qquad \delta_{m_0} = \frac{\beta^{2m_0+2}}{(2m_0+2)!}, \qquad \widetilde{P}_{m_0}(x) = \frac{P_{m_0}(x)}{1+\delta_{m_0}}.$$

The polynomial P_{m_0} is the Maclaurin polynomial of $\cos(\beta x)$ through degree $2m_0 + 1$ (the odd coefficient vanishes). Taylor's theorem and the bound on every derivative of cosine give $|P_{m_0}(x) - \cos(\beta x)| \leq \delta_{m_0}$ for $|x| \leq 1$. Hence $|\widetilde{P}_{m_0}(x)| \leq 1$ on $[-1, 1]$ and $\|\widetilde{P}_{m_0} - \cos(\beta\,\cdot)\|_\infty \leq 2\delta_{m_0}/(1+\delta_{m_0})$;

$\widetilde{P}_{m_0}$ is even of degree $2m_0$. It is therefore within the standard bounded-polynomial Hermitian-QSVT class for a block-encoding of G [2]. This establishes an admissible polynomial response, not a phase sequence, gate synthesis, or a compiled circuit. As a polynomial in the singular value σ/α, the response $\widetilde{P}_{m_0}((\sigma/\alpha)^2)$ has degree $4m_0$.

If a block-encoding of B is available, a product construction can encode $G = B^\dagger B$ using two calls to the B encoding, up to ancilla and reflection costs. We write d_G for the actual oracle-query cost of one implemented cosine-polynomial transform of G, and c_ψ for state preparation. Neither is numerically estimated here. A coherent implementation of t transforms has query depth proportional to td_G; a hard depth cap therefore constrains t as well as the polynomial degree. The normalization bound α and its acquisition cost are inputs, not free outputs of this method.

4 Finite-spectrum convergence and readout

Let $T = \{i : \lambda_i = \lambda_0\}$ be the minimum-eigenvalue eigenspace, with initial mass $a = \sum_{i\in T} p_i > 0$. Let $q_* = \cos(\beta\lambda_0)$ and suppose

$$q_i \le r q_* \quad (i \notin T), \qquad 0 \le r < 1.$$

For the cosine-Gram response, if an eigenvalue strictly larger than λ_0 exists, define $\lambda_+ = \min\{\lambda_i : \lambda_i > \lambda_0\}$; then $r = \cos(\beta\lambda_+)/\cos(\beta\lambda_0) < 1$. If no such eigenvalue exists, all spectral weight is already in the target subspace and $F_t = 1$.

Theorem 1 (Conditional spectral leakage)**.** *For every integer $t \ge 0$, let F_t be the probability mass of T in the normalized state $C^t|\psi\rangle/\sqrt{Z_t}$. Then*

$$1 - F_t \le \frac{1-a}{a} r^{2t}.$$

If $\mu_t = \langle C^t\psi|G|C^t\psi\rangle/Z_t$, then

$$0 \le \mu_t - \lambda_0 \le (1-\lambda_0)(1-F_t) \le \frac{1-a}{a} r^{2t}.$$

The proof is given in Appendix A. Lean formalization is in progress; Appendix B will state its exact scope and checker status. The identification of the finite weights with a matrix eigenbasis, the block-encoding implementation, and the readout interpretation are finite-dimensional mathematical derivations, not part of the scalar Lean theorem.

Finite-degree transfer condition. The leakage theorem also applies to a bounded polynomial response $\widetilde{q}_i = \widetilde{P}(\lambda_i)$, but boundedness alone does not ensure that it orders the target eigenvalue correctly. Let $q_{\text{out}} = \max_{i\notin T} q_i$ and let E be a uniform error bound $|\widetilde{P}(x) - \cos(\beta x)| \le E$ on $[0,1]$. If $E < (q_* - q_{\text{out}})/2$, then $|\widetilde{q}_*| \ge q_* - E > 0$ and $|\widetilde{q}_i|/|\widetilde{q}_*| \le (q_{\text{out}} + E)/(q_* - E) < 1$. The generic finite-weight bound therefore transfers with this ratio. This sufficient condition can be conservative and is not evaluated as a degree-optimal filter here.

Maximum-singular-value branch. For the power method, take response $q_i = \lambda_i$ and target the maximum eigenspace $T_{\max} = \{i : \lambda_i = \lambda_{\max}\}$. If its initial mass is $a_{\max} > 0$, $\lambda_{\max} > 0$, and $\lambda_- = \max\{\lambda_i : \lambda_i < \lambda_{\max}\}$ exists, the same finite-weight inequality applies with $q_* = \lambda_{\max}$ and $r = \lambda_-/\lambda_{\max}$. Its target leakage is at most $((1-a_{\max})/a_{\max})r^{2t}$, and its filtered mean obeys $0 \le \lambda_{\max} - \mu_t \le (1-F_t) \le ((1-a_{\max})/a_{\max})r^{2t}$. The maximum singular value estimate is $\alpha\sqrt{\widehat{\mu}_t}$

with corresponding error at most $\alpha\sqrt{((1-a_{\max})/a_{\max})r^{2t}+\zeta}$. If $\lambda_{\max} < 1$, the unnormalized success probability is at most $\lambda_{\max}^{2t}$; this concentration step has a success cost as well.

Suppose a block-encoding of B is used to measure the success flag without postselecting on it. On an input $|\psi_t\rangle$, that flag is Bernoulli with mean $\langle\psi_t|B^\dagger B|\psi_t\rangle = \mu_t$. From M accepted filtered-state preparations, Hoeffding's inequality gives

$$|\widehat{\mu}_t - \mu_t| \le \zeta \quad \text{with probability at least } 1-\delta \quad \text{if} \quad M \ge \frac{\log(2/\delta)}{2\zeta^2}.$$

Clip $\widehat{\mu}_t$ to $[0,1]$ and estimate $\sigma_0(A)$ by $\widehat{\sigma} = \alpha\sqrt{\widehat{\mu}_t}$. Then

$$|\widehat{\sigma} - \sigma_0(A)| \le \alpha\sqrt{\frac{1-a}{a}r^{2t}+\zeta}.$$

This is an additive-error statement under an ideal Bernoulli readout model; it does not account for noise, state-preparation error, or an unknown α.

Success and cost accounting. For the exact ideal cosine response, the target component alone gives $Z_t \ge aq_*^{2t}$, while $q_i \le q_*$ gives $Z_t \le q_*^{2t}$. If each failed filter attempt is discarded and restarted from $|\psi\rangle$, the expected number of attempts per accepted copy is $1/Z_t$. For an implemented polynomial response $\widetilde{P}(G)$, the success probability is instead $\widetilde{Z}_t = \|\widetilde{P}(G)^t\psi\|^2$; a finite-degree cost estimate must use $\widetilde{Z}_t$, not silently substitute ideal Z_t. A query proxy for M accepted readout samples is

$$M\left(\frac{c_\psi + td_G}{\widetilde{Z}_t} + d_B\right),$$

where d_B is the cost of one Bernoulli readout using the B block-encoding. This proxy is symbolic; finite-degree experiments below report only idealized spectral response probabilities and do not measure $\widetilde{Z}_t$ from a compiled circuit. Amplitude amplification, coherent recycling, and compiler-level gate costs are outside this accounting. In particular, when $\lambda_0 > 0$, $q_* < 1$ and the success probability can decay exponentially in t; a small conditional leakage does not by itself imply low total cost.

5 Numerical results

We ran a deterministic matrix-level simulation using exact classical eigendecompositions of seven right Gram matrices and fixed initial spectral weights. For each case, the input state is defined in the computed right-eigenvector basis; the calculation propagates the exact spectral weights rather than simulating a quantum state vector or circuit. The figure reports the exact $\cos(\beta G)$ response at $\beta = 1.2$. A separate Bernoulli sample uses seed 20261026 and 2,000 accepted spectral-model draws. The script and full JSON output are in `experiments/qmsve_matrix_level.py` and `results/qmsve_matrix_level.json`.

Case	$\sigma_{\min}$	$a_{\min}$	G^8 estimate	$\cos(\beta G)^8$ estimate / mass	$\cos(\beta\sqrt{G})^8$ estimate / mass
Square, separated	0.12	0.40	0.8600	0.2896 / 0.6791	0.1643 / 0.9417
Near-degenerate	0.20	0.30	0.8000	0.2076 / 0.4987	0.2027 / 0.5059
Degenerate minimum	0.20	0.40	0.7000	0.2713 / 0.9254	0.2041 / 0.9963
Zero target overlap	0.12	0	0.8600	0.4809 / 0	0.4803 / 0
Wide, full row rank	0	0.16	0.8000	0.2744 / 0.5666	0.1442 / 0.8715
Rank-deficient square	0	0.30	0.8000	0.2654 / 0.5714	0.1435 / 0.8719
Rotated tall, full column rank	0.15	0.35	0.6500	0.3065 / 0.8214	0.1630 / 0.9898

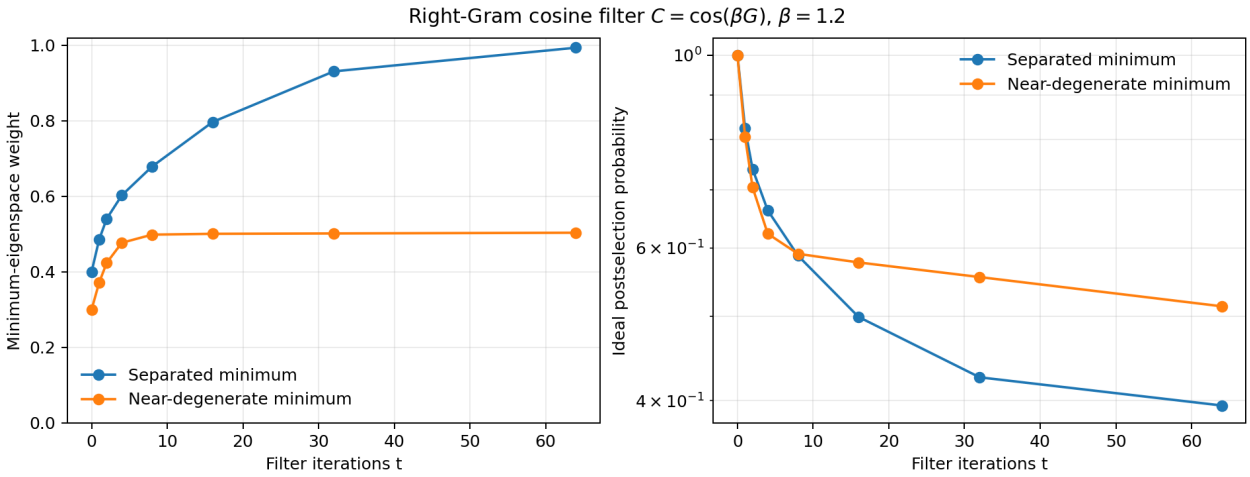


Figure 1: Exact matrix-level conditional target weight and ideal filter success probability. The separated case concentrates on the minimum eigenspace; the near-degenerate case does not do so over the displayed iteration range.

The G^8 column reports the estimated largest singular value; the cosine columns report the estimated smallest singular value and conditional minimum-eigenspace mass. For the square-gap case, G^8 has ideal success probability 0.00200, whereas the cosine-Gram response has success probability 0.5876. This distinction prevents a conditional estimate from being mistaken for a low-cost accepted sample.

For the square-gap example, finite-degree Taylor responses at $t = 8$ give:

Degree in G	Degree in σ/α	Uniform error bound	$\widehat{\sigma}$	Target mass	Ideal success probability
2	4	1.591×10^{-1}	0.2891	0.6801	0.1558
4	8	8.260×10^{-3}	0.2896	0.6791	0.5499
6	12	2.133×10^{-4}	0.2896	0.6791	0.5866
12	24	2.946×10^{-10}	0.2896	0.6791	0.5876

These are polynomial-response evaluations with a global analytic boundedness margin. They are not phase-angle synthesis results; the table reports the idealized spectral success probability, not a compiled-circuit success measurement.

At $t = 64$, the separated spectrum $(0.12, 0.48, 0.86)$ has minimum-eigenspace mass 0.994 and ideal filter success probability 0.395 under the original cosine-Gram response. The near-degenerate case $(0.20, 0.205, 0.80)$ reaches only mass 0.504 with success probability 0.514. A degenerate minimum with initial subspace mass 0.40 reaches mass greater than 0.999999; the method selects the eigenspace, not a unique vector. If the initial state has zero mass on the minimum eigenspace, the filter never creates that component: in the tested instance the estimate remains at 0.48 while the ideal success probability falls to 0.00282.

The matrix-level model also shows why a right-space definition matters. A wide full-row-rank matrix has a right-kernel eigenvalue zero in G; the method targets that kernel rather than the smallest nonzero singular value. The rank-deficient square example likewise converges toward its zero singular value. A normalization check rejects a matrix whose norm exceeds the declared $\alpha = 1$.

For the separated case at $t = 8$, the exact filtered Gram mean is 0.08385. The 2,000-shot Bernoulli estimate is 0.0780, within the stated 95% Hoeffding radius 0.03037. The expected number of fresh filter attempts for these accepted samples is approximately 3,404 under the ideal model. This is a scalar shot-noise check, not a circuit experiment.

The distinct dilation response $\cos(\beta\sigma)$ concentrates faster on the separated example (target mass 0.996 at $t = 16$ versus 0.797 for $\cos(\beta\sigma^2)$), but its success probability is lower (0.288 versus 0.500). For the near-degenerate example both responses remain close to 0.5 at $t = 16$. These numbers compare exact spectral responses only: their polynomial degrees, encodings, and compiled costs are not matched.

6 Discussion and limitations

The finite-spectrum argument isolates the mechanism behind the triangular iteration: a strictly largest filter response on the target eigenspace yields geometric suppression of all other spectral weight. The same calculation applies to other diagonalizable filters under the same response-ratio condition, so the leakage inequality itself should be read as an elementary filter bound, not as a new quantum algorithmic primitive. The practical question is whether a chosen response can be implemented at a useful polynomial degree and whether enough accepted samples can be obtained within the available depth and query budget. This report does not answer that end-to-end optimization problem.

The result has several explicit assumptions. The target component must have nonzero initial mass; a useful guarantee requires a quantitative lower bound on that mass. A strict response ratio $r < 1$ is needed, which for the cosine-Gram filter follows from a nonzero spectral gap above the minimum eigenvalue. Repeated postselection incurs expected overhead $1/Z_t$, and the success probability can shrink exponentially with t when the target eigenvalue is positive. The normalization α must be known and its acquisition cost is not included. The polynomial Taylor proxy is bounded on the assumed block-encoding spectrum, but no QSVT phases are generated and no finite-precision or synthesis error is propagated through t iterations. The readout analysis assumes ideal independent Bernoulli trials and a prepared accepted state.

The numerical examples are deliberately small and chosen to expose degeneracy, small gaps, zero overlap, and kernel effects. They do not establish performance on chemistry instances, large sparse matrices, or hardware. In particular, no speedup, improved query complexity, or novelty over existing power methods and polynomial filtering is claimed. A resource comparison would need to fix a common input model and include state preparation, normalization, block-encoding, filter degree, success amplification or restart, readout shots, and finite-precision errors.

7 Conclusion

We gave an assumption-explicit formulation of a cosine-based minimum-singular-value iteration on the normalized right Gram operator and stated its conditional finite-spectrum convergence and Bernoulli readout guarantees. We separated the $\cos(\beta\sigma^2/\alpha^2)$ Gram response from a distinct $\cos(\beta\sigma/\alpha)$ response, and audited both on seven small spectral examples. The examples confirm the role of gap, target overlap, degeneracy, and right-kernel support. They also show why conditional convergence alone is insufficient to establish an efficient algorithm: success probability and implementation cost must be counted. Establishing a useful end-to-end resource advantage remains open in this study.

A Proof of the finite-spectrum bounds

Write $Q_T = \sum_{i\in T} p_i q_*^{2t}$ for the unnormalized target weight and $Q_\perp = \sum_{i\notin T} p_i q_i^{2t}$. The assumed response bound implies

$$Q_\perp \le r^{2t} q_*^{2t} \sum_{i\notin T} p_i = (1-a) r^{2t} q_*^{2t}, \qquad Q_T = a q_*^{2t}.$$

Since $F_t = Q_T/(Q_T + Q_\perp)$, we have

$$1 - F_t = \frac{Q_\perp}{Q_T + Q_\perp} \le \frac{Q_\perp}{Q_T} \le \frac{1-a}{a} r^{2t}.$$

Every eigenvalue outside T lies in $[\lambda_0, 1]$, so the filtered mean satisfies

$$0 \le \mu_t - \lambda_0 = \sum_{i\notin T} \frac{p_i q_i^{2t}}{Z_t} (\lambda_i - \lambda_0) \le (1 - F_t)(1 - \lambda_0) \le \frac{1-a}{a} r^{2t}.$$

For a readout estimate with $|\widehat{\mu}_t - \mu_t| \le \zeta$, clipping to $[0,1]$ cannot increase its distance from $\lambda_0 \in [0,1]$. Thus $|\widehat{\mu}_t - \lambda_0| \le \mu_t - \lambda_0 + \zeta$. For nonnegative x, y, $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$, giving the stated singular-value error bound after multiplication by α.

B Lean formalization status

The exact finite-weight postselection inequality and its posterior-moment consequence are checked in Lean 4. The formal source is reproduced in Appendix D; it was checked with Lean 4.35.0 and Mathlib v4.35.0 using `lake build QMSVE` and `lake env lean Audit.lean`, both with exit code 0. The matrix spectral decomposition, identification of filter weights with matrix eigencomponents, QSVT implementation, and Bernoulli concentration remain outside the scalar formalization.

C Source manuscript metadata

The QMSVE source used for this reassessment is the supplied manuscript titled *Quantum maximum/minimum singular-value estimation algorithms*. The copy examined does not state author, venue, or publication metadata, and labels the triangular iteration as provisional. The manuscript's author field is consequently left as a placeholder in the main file; no identity or affiliation is inferred.

D Lean source code

The following is the complete formal source used for the two scalar statements above. Its imports are pinned by the adjacent Lean project files.

```
import Mathlib

/-!
Finite scalar power-iteration bound used in the QMSVE appendix.

The proof starts at the spectral-weight reduction: `w` are the normalized
initial spectral weights and `q` are nonnegative scalar response values.
Matrix/SVD and QSVT realization are outside this formal statement.
-/

theorem weighted_power_postselection_bound
    {ι : Type*} [Fintype ι] [DecidableEq ι]
    (w q : ι → ℝ) (T : Finset ι) (qStar r : ℝ) (t : ℕ)
    (hw_nonneg : ∀ i, 0 ≤ w i)
```

```
  (hw_sum : ∑ i, w i = 1)
  (hq_nonneg : ∀ i, 0 ≤ q i)
  (hqStar_pos : 0 < qStar)
  (hr_nonneg : 0 ≤ r)
  (hr_lt_one : r < 1)
  (hq_target : ∀ i ∈ T, q i = qStar)
  (hq_outside : ∀ i ∉ T, q i ≤ r * qStar)
  (hmass_pos : 0 < ∑ i ∈ T, w i) :
  let a : ℝ := ∑ i ∈ T, w i
  let m : ℕ := 2 * t
  let qPow : ℝ := qStar ^ m
  let P : ℝ := ∑ i, w i * q i ^ m
  0 < P ∧ a * qPow ≤ P ∧ P ≤ qPow ∧
    1 - (a * qPow) / P ≤ ((1 - a) / a) * r ^ m := by
dsimp only
let a : ℝ := ∑ i ∈ T, w i
let m : ℕ := 2 * t
let qPow : ℝ := qStar ^ m
let P : ℝ := ∑ i, w i * q i ^ m
let E : ℝ := ∑ i ∈ Tᶜ, w i * q i ^ m

have hqStar_nonneg : 0 ≤ qStar := le_of_lt hqStar_pos
have hr_le_one : r ≤ 1 := le_of_lt hr_lt_one
have hscale : r * qStar ≤ qStar := by
  simpa using (mul_le_mul_of_nonneg_right hr_le_one hqStar_nonneg)

have hsplitW : a + (∑ i ∈ Tᶜ, w i) = 1 := by
  calc
    a + (∑ i ∈ Tᶜ, w i) = (∑ i ∈ T, w i) + ∑ i ∈ Tᶜ, w i := by rfl
    _ = ∑ i, w i := Finset.sum_add_sum_compl T w
    _ = 1 := hw_sum
have hcompW_nonneg : 0 ≤ ∑ i ∈ Tᶜ, w i :=
  Finset.sum_nonneg fun i hi => hw_nonneg i
have ha_le_one : a ≤ 1 := by linarith
have hcompW : (∑ i ∈ Tᶜ, w i) = 1 - a := by linarith

have hq_le_star : ∀ i, q i ≤ qStar := by
  intro i
  by_cases hi : i ∈ T
  · rw [hq_target i hi]
  · exact (hq_outside i hi).trans hscale

have htargetP : (∑ i ∈ T, w i * q i ^ m) = a * qPow := by
  calc
    (∑ i ∈ T, w i * q i ^ m) = ∑ i ∈ T, w i * qPow := by
      apply Finset.sum_congr rfl
      intro i hi
      rw [hq_target i hi]
    _ = (∑ i ∈ T, w i) * qPow := (Finset.sum_mul T w qPow).symm
    _ = a * qPow := by rfl

have hdecomp : a * qPow + E = P := by
  have h := Finset.sum_add_sum_compl T (fun i => w i * q i ^ m)
  rw [htargetP] at h
  simpa [P, E] using h

have hE_nonneg : 0 ≤ E := by
  apply Finset.sum_nonneg
  intro i hi
  exact mul_nonneg (hw_nonneg i) (pow_nonneg (hq_nonneg i) m)

have htarget_pos : 0 < a * qPow :=
  mul_pos hmass_pos (pow_pos hqStar_pos m)
have hqPow_pos : 0 < qPow := pow_pos hqStar_pos m
have hP_lower : a * qPow ≤ P := by linarith [hdecomp, hE_nonneg]
have hP_pos : 0 < P := lt_of_lt_of_le htarget_pos hP_lower

have hP_upper : P ≤ qPow := by
  calc
    P = ∑ i, w i * q i ^ m := rfl
    _ ≤ ∑ i, w i * qPow := by
      apply Finset.sum_le_sum
      intro i hi
      exact mul_le_mul_of_nonneg_left
        (pow_le_pow_left₀ (hq_nonneg i) (hq_le_star i) m) (hw_nonneg i)
    _ = (∑ i, w i) * qPow := (Finset.sum_mul Finset.univ w qPow).symm
    _ = qPow := by rw [hw_sum]; simp

have hE_upper : E ≤ (1 - a) * r ^ m * qPow := by
  calc
    E = ∑ i ∈ Tᶜ, w i * q i ^ m := rfl
    _ ≤ ∑ i ∈ Tᶜ, w i * (r * qStar) ^ m := by
      apply Finset.sum_le_sum
      intro i hi
      exact mul_le_mul_of_nonneg_left
        (pow_le_pow_left₀ (hq_nonneg i) (hq_outside i (Finset.mem_compl.mp hi)) m)
        (hw_nonneg i)
    _ = (∑ i ∈ Tᶜ, w i) * (r * qStar) ^ m :=
      (Finset.sum_mul Tᶜ w ((r * qStar) ^ m)).symm
    _ = (1 - a) * r ^ m * qPow := by
      rw [hcompW, mul_pow]
      simp [qPow]
```

```
      ring

  have hEbound_nonneg : 0 ≤ (1 - a) * r ^ m * qPow :=
    mul_nonneg (mul_nonneg (sub_nonneg.mpr ha_le_one) (pow_nonneg hr_nonneg m))
      (pow_nonneg hqStar_nonneg m)
  have hquot : ((1 - a) * r ^ m * qPow) / (a * qPow) =
      ((1 - a) / a) * r ^ m := by
    calc
      ((1 - a) * r ^ m * qPow) / (a * qPow) = ((1 - a) * r ^ m) / a := by
        simpa [mul_assoc] using
          (mul_div_mul_right (c := qPow) ((1 - a) * r ^ m) a
            (ne_of_gt hqPow_pos))
      _ = ((1 - a) / a) * r ^ m := by ring
  have hEdiv : E / P ≤ ((1 - a) * r ^ m * qPow) / P :=
    div_le_div_of_nonneg_right hE_upper (le_of_lt hP_pos)
  have hbounddiv : ((1 - a) * r ^ m * qPow) / P ≤
      ((1 - a) * r ^ m * qPow) / (a * qPow) :=
    div_le_div_of_nonneg_left hEbound_nonneg htarget_pos hP_lower
  have hleak_eq : 1 - (a * qPow) / P = E / P := by
    rw [← hdecomp]
    field_simp [ne_of_gt hP_pos]
    ring

  exact ⟨hP_pos, hP_lower, hP_upper,
    by rw [hleak_eq]; calc
      E / P ≤ ((1 - a) * r ^ m * qPow) / P := hEdiv
      _ ≤ ((1 - a) * r ^ m * qPow) / (a * qPow) := hbounddiv
      _ = ((1 - a) / a) * r ^ m := hquot⟩

/-!
If the postselected spectral weights are `v`, the scalar moment differs from
the selected extremal eigenvalue by at most the unselected mass.  This is the
finite-sum readout step; measurement concentration is not part of the theorem.
-/

theorem posterior_moment_deviation_bound
    {ι : Type*} [Fintype ι] [DecidableEq ι]
    (v lam : ι → ℝ) (T : Finset ι) (lambda0 : ℝ)
    (hv_nonneg : ∀ i, 0 ≤ v i)
    (hv_sum : ∑ i, v i = 1)
    (hlambda0_nonneg : 0 ≤ lambda0)
    (hlam_bounds : ∀ i, lambda0 ≤ lam i ∧ lam i ≤ 1)
    (hlam_target : ∀ i ∈ T, lam i = lambda0) :
    let F : ℝ := ∑ i ∈ T, v i
    let μ : ℝ := ∑ i, v i * lam i
    0 ≤ μ - lambda0 ∧ μ - lambda0 ≤ (1 - lambda0) * (1 - F) := by
  dsimp only
  let F : ℝ := ∑ i ∈ T, v i
  let μ : ℝ := ∑ i, v i * lam i
  let Vout : ℝ := ∑ i ∈ Tᶜ, v i
  let Mout : ℝ := ∑ i ∈ Tᶜ, v i * lam i

  have hFsplit : F + Vout = 1 := by
    calc
      F + Vout = (∑ i ∈ T, v i) + ∑ i ∈ Tᶜ, v i := by rfl
      _ = ∑ i, v i := Finset.sum_add_sum_compl T v
      _ = 1 := hv_sum

  have hMsplit : lambda0 * F + Mout = μ := by
    have hTarget : (∑ i ∈ T, v i * lam i) = lambda0 * F := by
      calc
        (∑ i ∈ T, v i * lam i) = ∑ i ∈ T, v i * lambda0 := by
          apply Finset.sum_congr rfl
          intro i hi
          rw [hlam_target i hi]
        _ = (∑ i ∈ T, v i) * lambda0 := (Finset.sum_mul T v lambda0).symm
        _ = lambda0 * F := by dsimp [F]; ring
    have h := Finset.sum_add_sum_compl T (fun i => v i * lam i)
    rw [hTarget] at h
    simpa [μ, Mout] using h

  have hGapEq : μ - lambda0 = ∑ i ∈ Tᶜ, v i * (lam i - lambda0) := by
    calc
      μ - lambda0 = (lambda0 * F + Mout) - lambda0 := by rw [hMsplit]
      _ = Mout - lambda0 * Vout := by linear_combination lambda0 * hFsplit
      _ = (∑ i ∈ Tᶜ, v i * lam i) - lambda0 * (∑ i ∈ Tᶜ, v i) := by
        simp [Mout, Vout]
      _ = (∑ i ∈ Tᶜ, v i * lam i) - ∑ i ∈ Tᶜ, v i * lambda0 := by
        rw [← Finset.sum_mul]
        ring
      _ = ∑ i ∈ Tᶜ, (v i * lam i - v i * lambda0) :=
        (Finset.sum_sub_distrib (s := Tᶜ) (fun i => v i * lam i)
          (fun i => v i * lambda0)).symm
      _ = ∑ i ∈ Tᶜ, v i * (lam i - lambda0) := by
        apply Finset.sum_congr rfl
        intro i hi
        ring

  have hGap_nonneg : 0 ≤ ∑ i ∈ Tᶜ, v i * (lam i - lambda0) := by
    apply Finset.sum_nonneg
    intro i hi
    exact mul_nonneg (hv_nonneg i) (sub_nonneg.mpr (hlam_bounds i).1)
```

```
have hGap_upper : (∑ i ∈ Tᶜ, v i * (lam i - lambda0)) ≤
    (1 - lambda0) * (1 - F) := by
  calc
    (∑ i ∈ Tᶜ, v i * (lam i - lambda0)) ≤ ∑ i ∈ Tᶜ, v i * (1 - lambda0) := by
      apply Finset.sum_le_sum
      intro i hi
      exact mul_le_mul_of_nonneg_left
        (sub_le_sub_right (hlam_bounds i).2 lambda0) (hv_nonneg i)
    _ = Vout * (1 - lambda0) := by
      change (∑ i ∈ Tᶜ, v i * (1 - lambda0)) =
        (∑ i ∈ Tᶜ, v i) * (1 - lambda0)
      exact (Finset.sum_mul Tᶜ v (1 - lambda0)).symm
    _ = (1 - lambda0) * (1 - F) := by rw [← hFsplit]; ring

exact ⟨by rw [hGapEq]; exact hGap_nonneg,
  by rw [hGapEq]; exact hGap_upper⟩
```

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